

Battery

Today, batteries are useful if you can't plug a device into an outlet, but when they were first invented, they were the only things capable of generating an electric current.

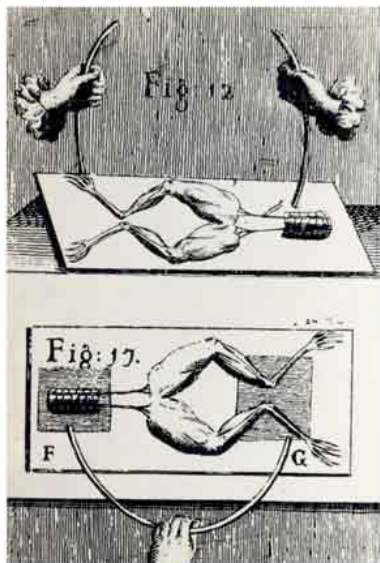
The invention that ELECTRIFIED the planet

Electrical storm

Electricity is as old as lightning. In 1752, American scientist **BENJAMIN FRANKLIN** flew a kite with a metal key in a storm and realized that the sparks coming from the metal proved that lightning is a *form of electricity*. This static ("at rest") electricity was first discovered in ancient Greece by a mathematician named Thales who produced a **static electric charge** by polishing amber with animal fur.



In Galvani's experiment, the frog was the equivalent of the brine-soaked paper in Volta's pile.



Frog findings

Even though people knew about electricity, they did not yet know how to produce an electric current—a flow of electric charge. In 1780, Italian doctor **LUIGI GALVANI** noticed that the muscles in the legs of dead frogs twitched when they made contact with two different metals. He thought this was caused by "*animal electricity*" in the frog itself. But fellow scientist **ALESSANDRO VOLTA** realized that the electricity wasn't due to the dead frog, but to the different metals.

Did you know?

French emperor Napoleon I was so impressed with Alessandro Volta's invention of the voltaic pile that he made him a count.

It paved the way for...

Understanding electric current led to the **ELECTRICITY SUPPLY** we have come to rely on today. The first public electricity lit the streets of Godalming, England, in **1881**.

